

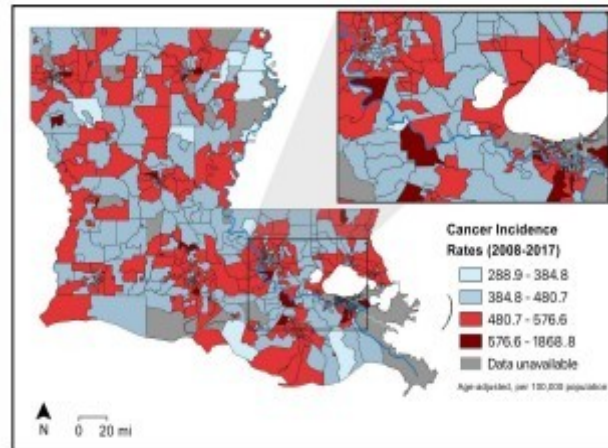
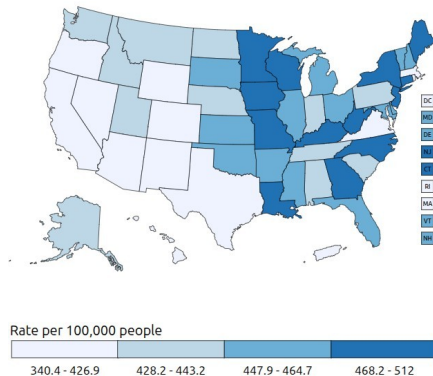
Industrialization and Cancer in Louisiana

Arturo Altamirano
CMPS 3160: Intro to Data Science

Motivations

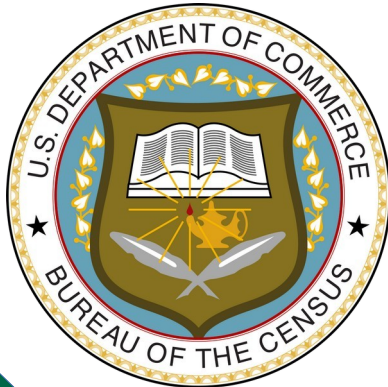
- ***“Air pollution is linked to higher cancer rates among black or impoverished communities in Louisiana”*** by Kimberly A Terrell and Gianna St Julien
- **Are these claims true? Can we visualize these claims and uphold them?**
- **Our analysis contributes to the growing body of evidence that Black and Brown communities in Louisiana are overburdened with the negative effects toxic air pollution from petrochemical facilities and other sources...**

Rate of New Cancers in the United States, 2022
All Cancer Sites Combined, Male and Female, All Races and Ethnicities



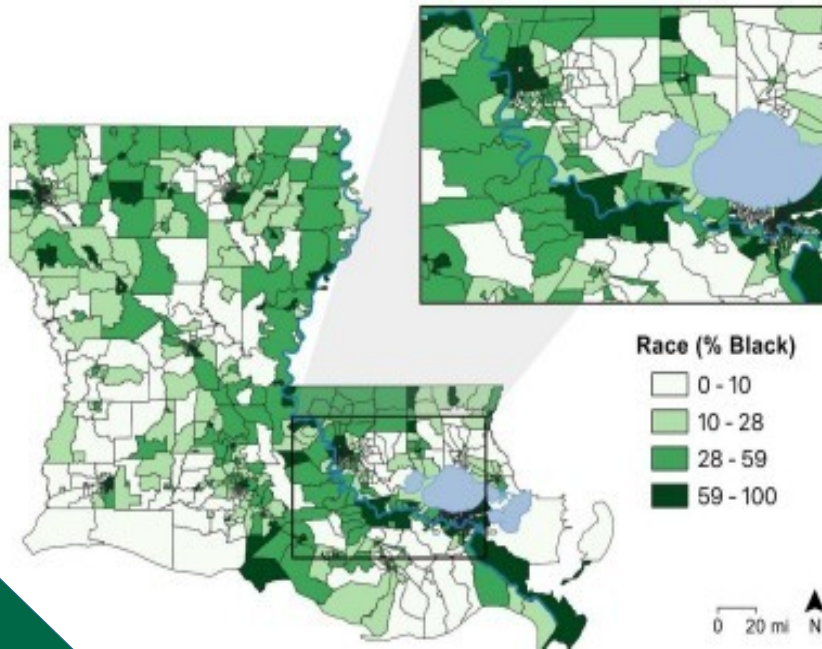
Data Collection and ETL

- Census Bureau's Community Survey
- Clinical Trials and Cancer Studies conducted by cancer.gov
- Accumulated for analysis on data.world

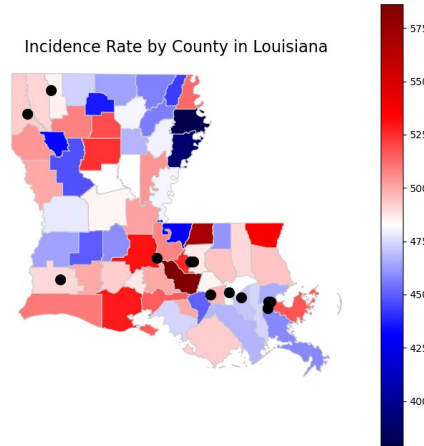


**NATIONAL
CANCER
INSTITUTE**

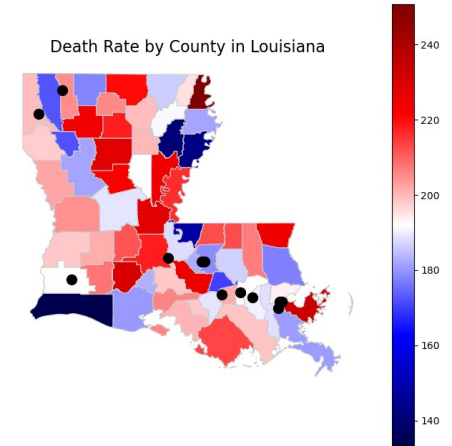
EDA – On The Right Track



Incidence Rate by County in Louisiana



Death Rate by County in Louisiana



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#points of petrochemical refineries - https://www.industryselect.com/blog/top-10-manufacturing-companies-in-louisiana
#it would be great if there was a dataset of these somewhere but I have been unable to find any
points_df = pd.DataFrame({
    'Latitude': [30.483988, 29.846726, 29.938748, 30.031041, 32.797383,
                30.065276, 30.526713, 30.240865, 29.933837, 29.995503, 30.476733, 32.470366],
    'Longitude': [-91.169601, -89.991966, -89.970822, -90.897768, -93.415004,
                 -90.593545, -91.746182, -93.272194, -89.944371, -90.409877, -91.210874, -93.789677],
    'Label': ['Baton Rouge Refinery', 'Alliance Refinery', 'Chalmette Refinery', 'Convent Refinery', 'Cotton Valley Refinery',
             'Garyville Refinery', 'Delek Refinery', 'Phillips 66 Lake Charles', 'Meraux Valero', 'Shell Norco',
             'Placid Refinery Port Allen', 'Calumet Shreveport']})

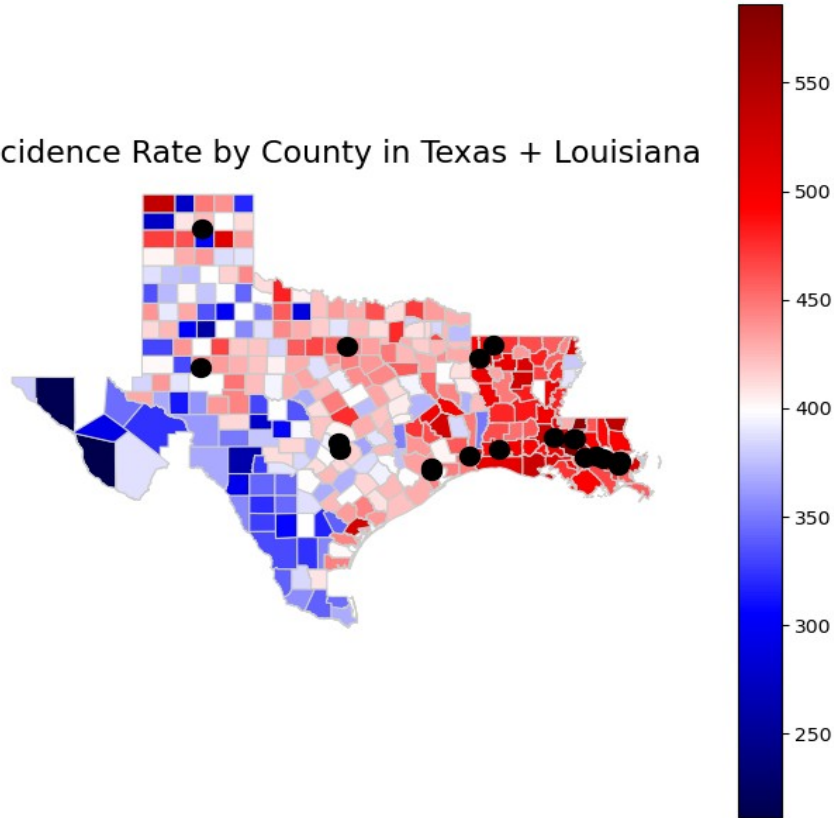
#convert lat/lon to Point geometry using shapely.point
geometry = [Point(xy) for xy in zip(points_df['Longitude'], points_df['Latitude'])]

#create another geodataframe out of these points
points_gdf = gpd.GeoDataFrame(points_df, geometry=geometry, crs='EPSG:4326')

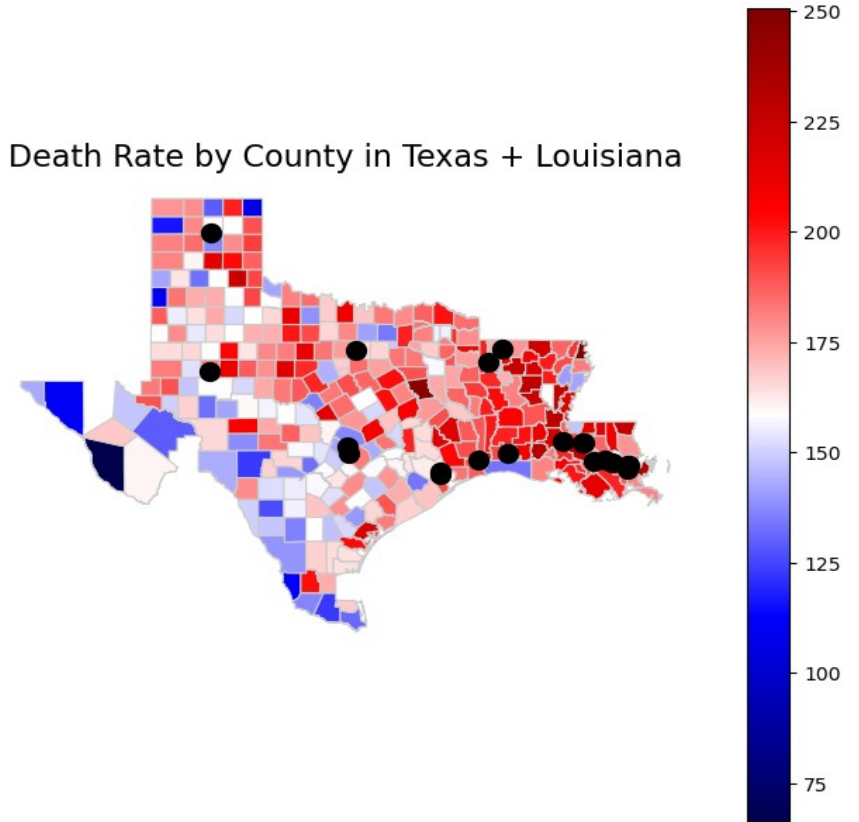
incidence_plotter(merged_Louisiana, 'Louisiana', points_gdf)
```

EDA - Expanded

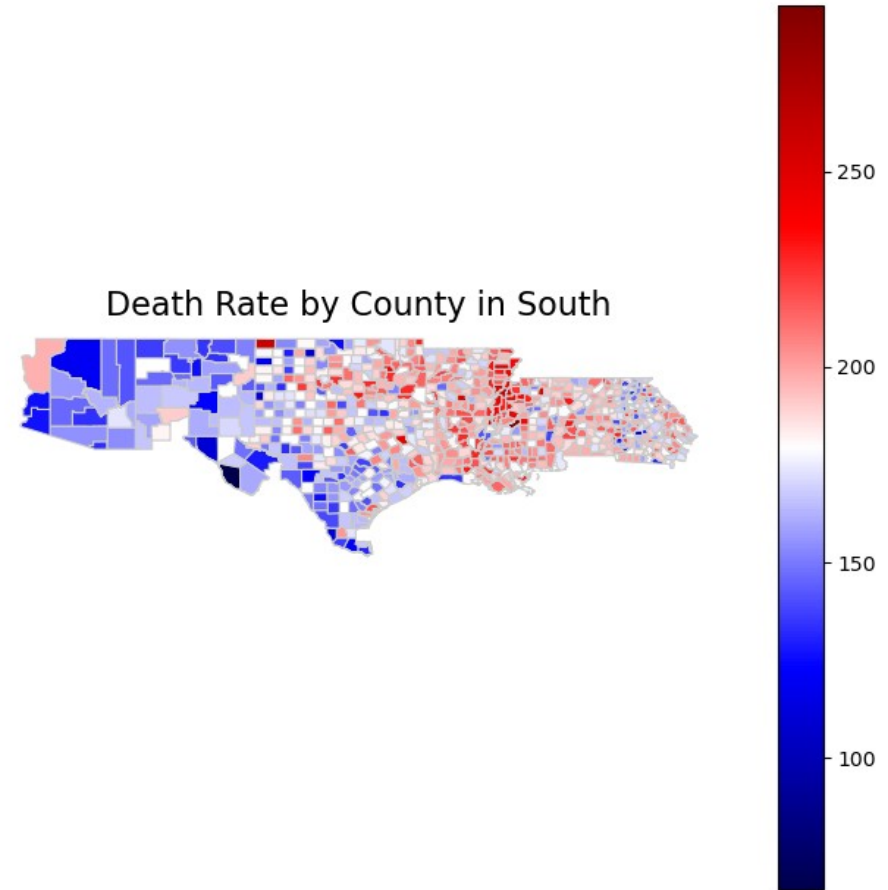
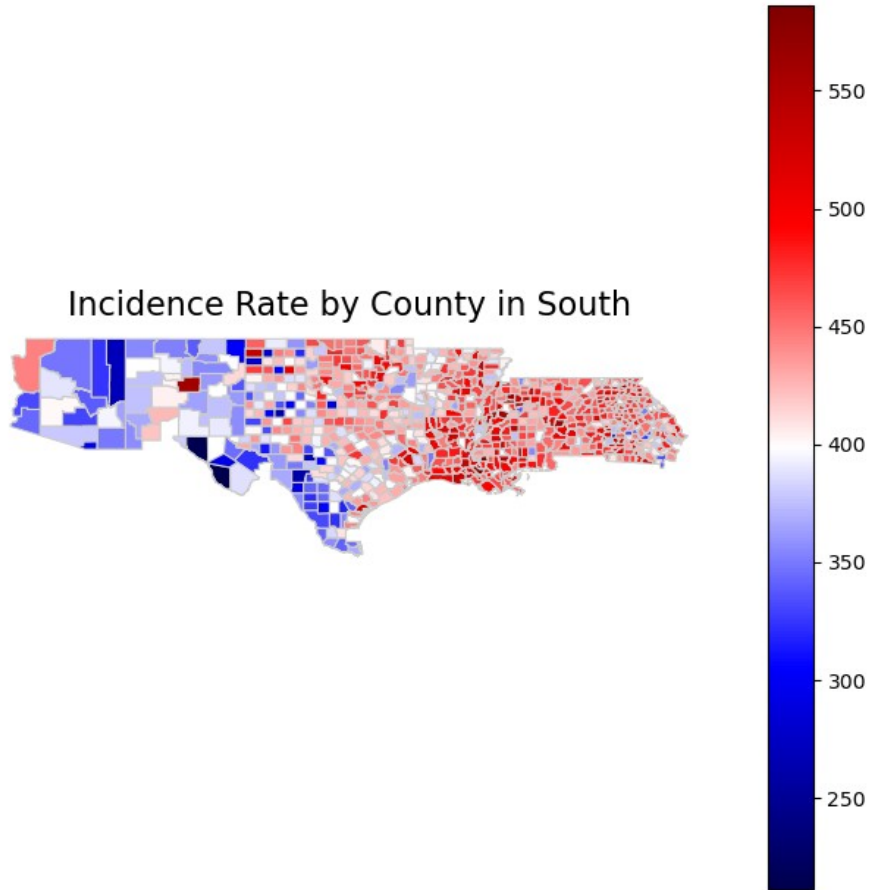
Incidence Rate by County in Texas + Louisiana



Death Rate by County in Texas + Louisiana

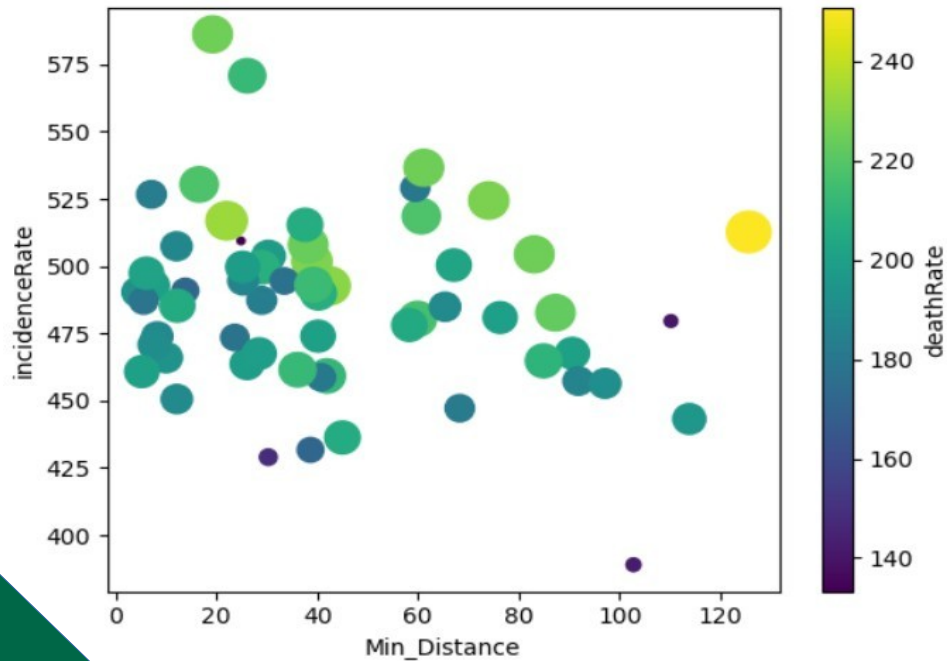


EDA - Expanded

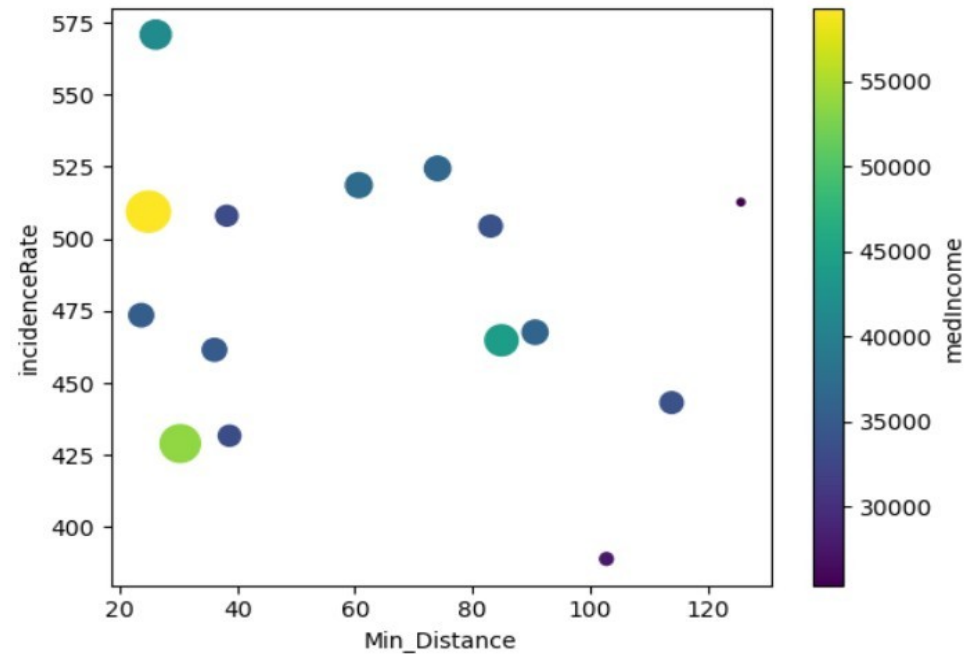


Correlation with Distance

Distance to Incidence Rate



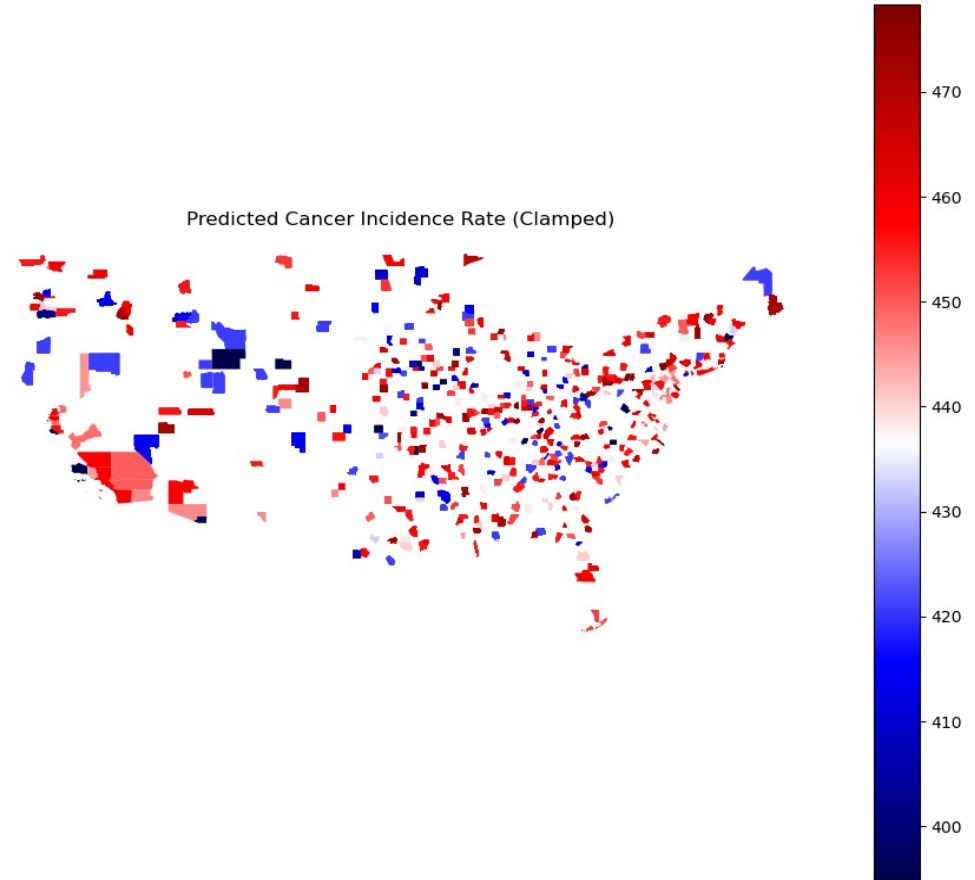
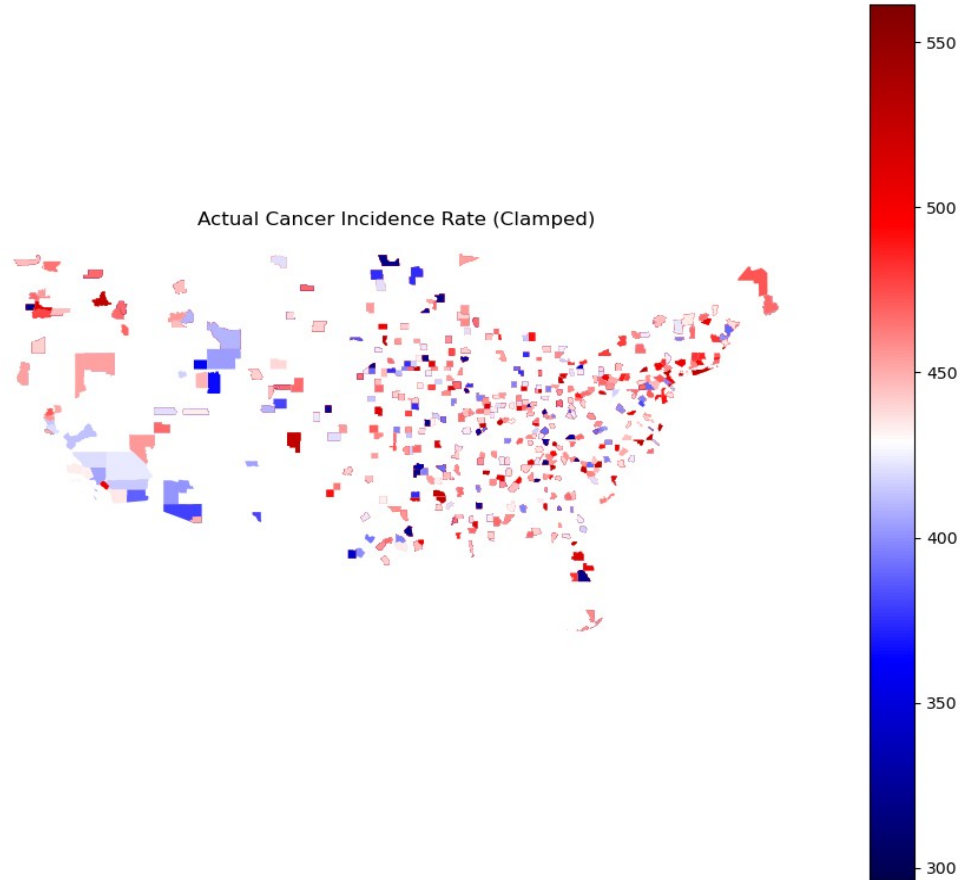
Distance to Incidence Rate:
Median Income below \$20,000



Model Training

- A 10NN regressor model yielded decent results when trained on all states recorded pollution metrics.
- Train on pollution levels, predict cancer incidence.

Minor Problem



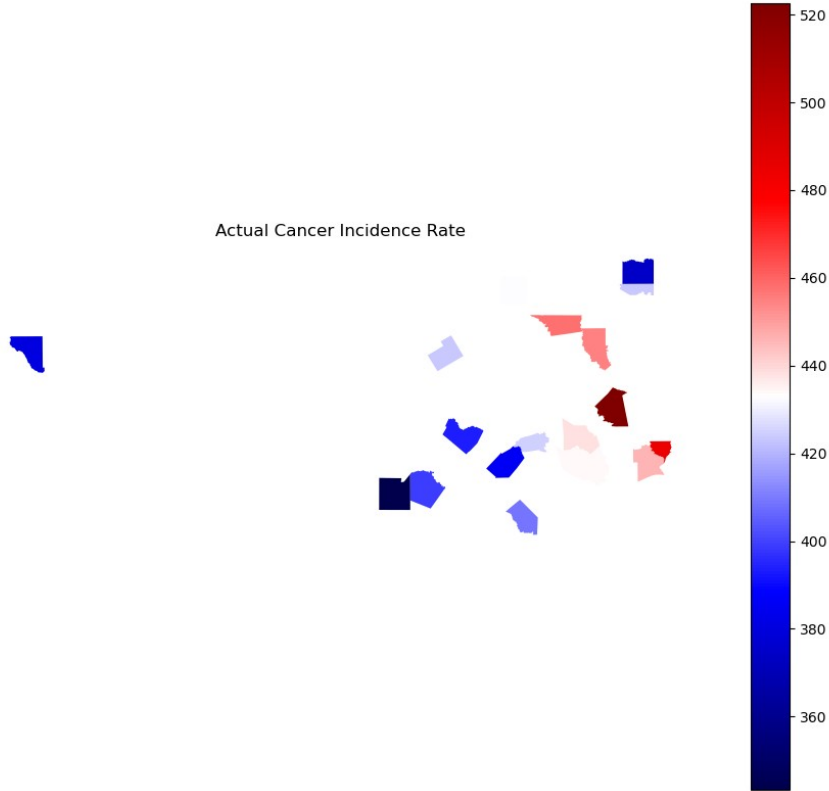
Model Results – California

The figure displays two maps of California, each showing cancer incidence rates by county. The left map, titled 'Actual Cancer Incidence Rate', uses a color scale from 390 (dark blue) to 470 (dark red). The right map, titled 'Predicted Cancer Incidence Rate (Model)', uses a color scale from 457 (dark blue) to 461 (dark red). Both maps show a high concentration of red in the northern and central regions, indicating higher incidence rates, and blue in the southern and coastal regions, indicating lower incidence rates. The predicted map shows a slightly different distribution of colors compared to the actual map, reflecting the model's output.

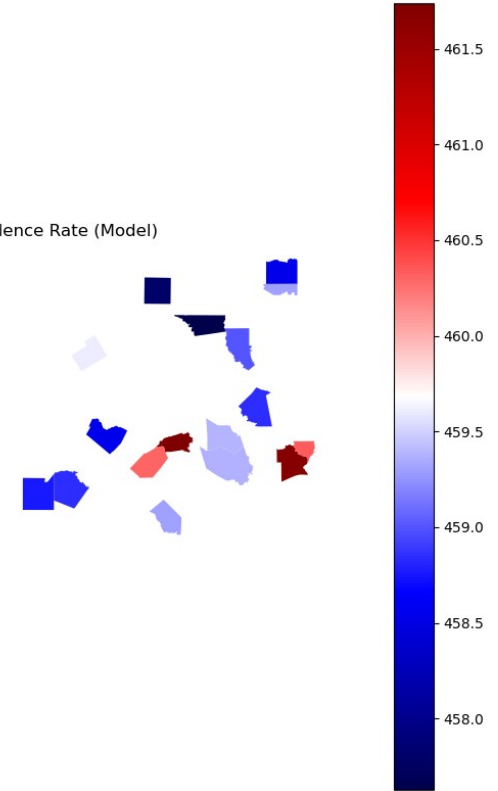


Model Results – Texas

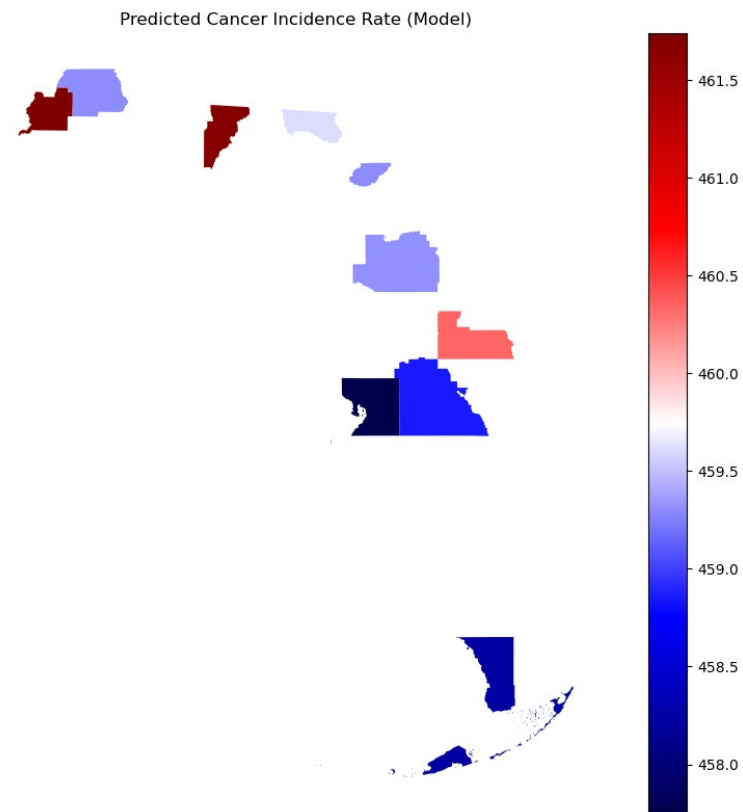
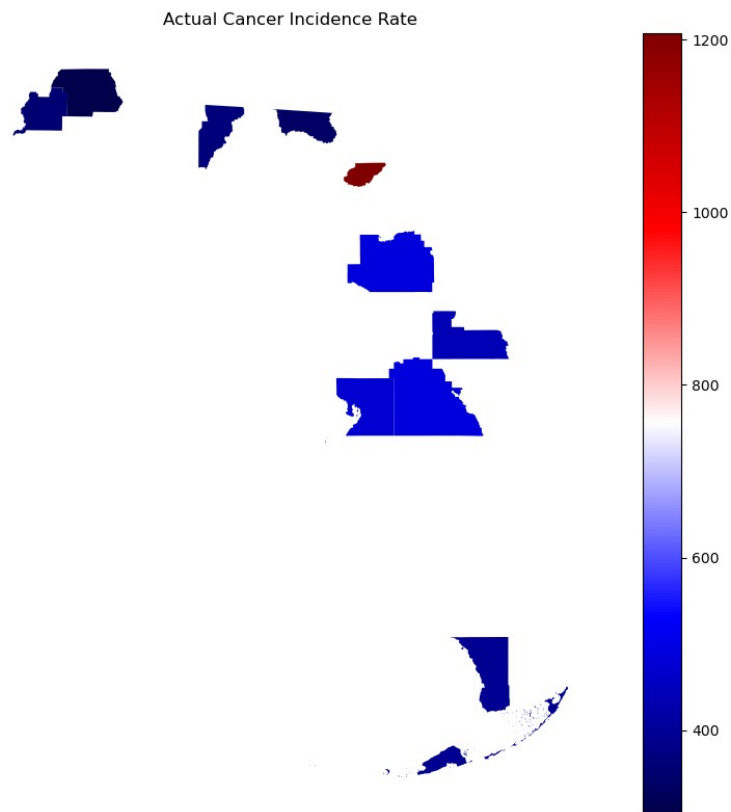
Actual Cancer Incidence Rate



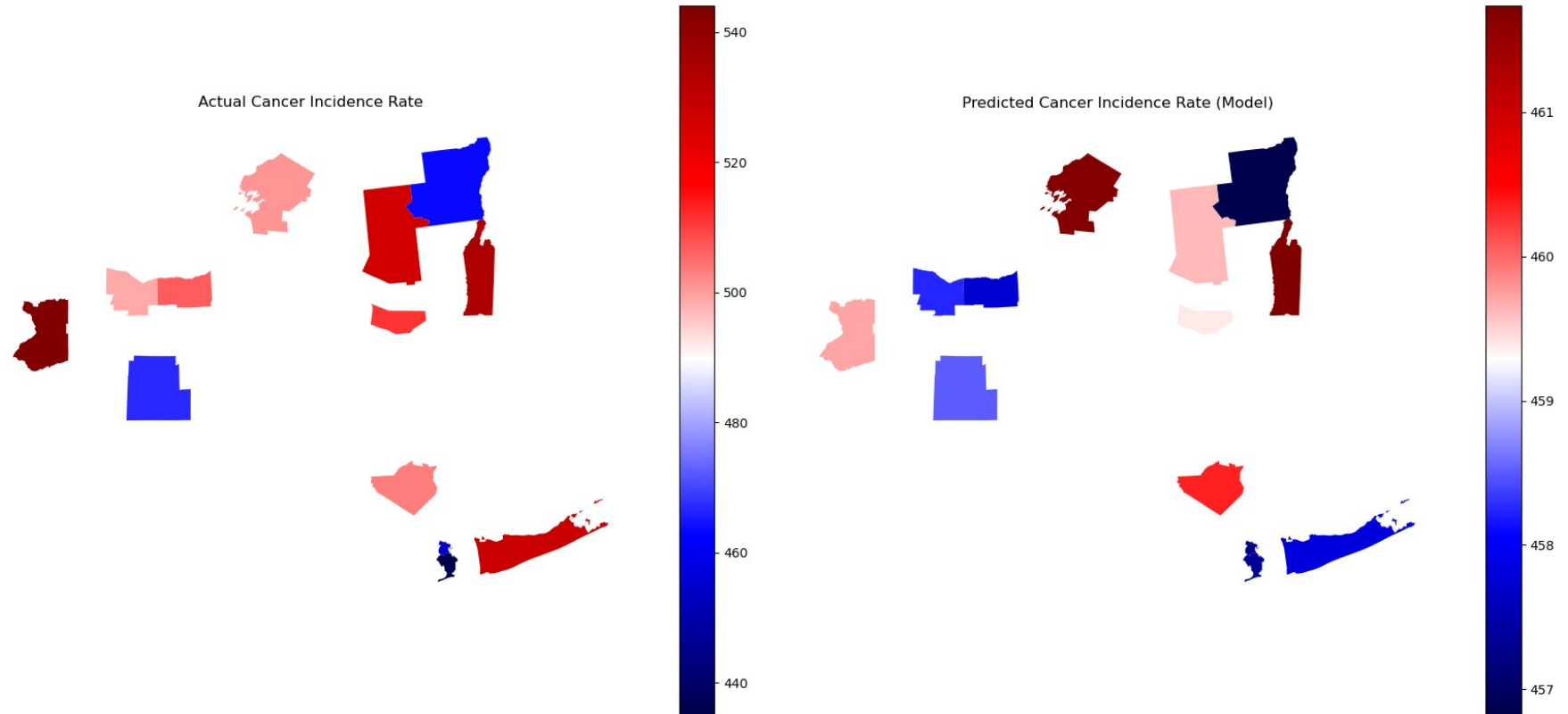
Predicted Cancer Incidence Rate (Model)



Model Results – Florida



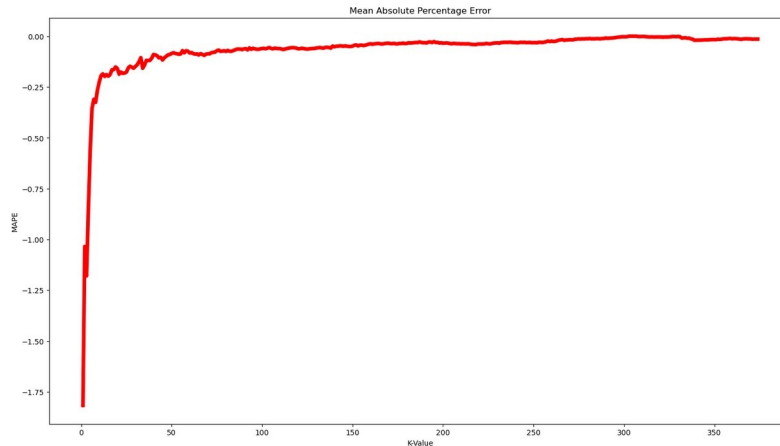
Model Results – New York



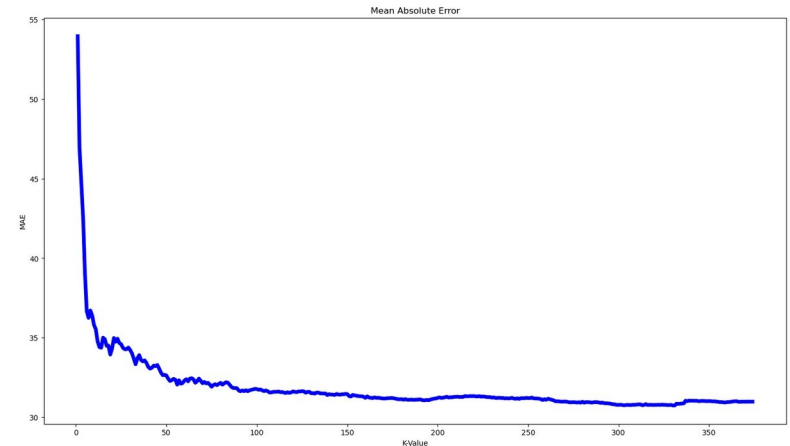
Parameter Tuning

- Need more data for such a complex relationship.
- Still see increasing performance.

R2 Score



Mean Absolute Error

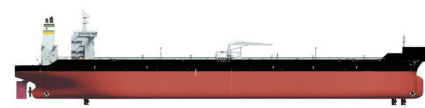


Model Results

- Single digit error rates on 80/20 train test split.
- Could see higher performance with more data, or more tuning.
- I would like to expand this dataset myself, with the distance to nearest plant metric as my pivot point.

Future Work

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Takeaways

- Noticeable concentration near petrochemical plants and the Mississippi river
- Findings of the Tulane paper are upheld
- Insights gained from this data can be used to accurately predict trends elsewhere